

# Case Study: Education, Ethnicity and Wages

## Afternoon Exercise Workbook

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## 1 Welcome to the Afternoon Session

This workbook guides you through a complete quantile regression analysis using real data. Work through the exercises in order, each one builds on the previous.

### The research question:

*Does the return to education differ between low and high earners? Is the ethnic wage gap uniform across the earnings distribution, or is it larger at the bottom?*

These are questions that OLS regression cannot fully answer. By the end of this session you will have answered both using quantile regression.

**Data:** CPS1988, 28,155 male workers from the US Current Population Survey, 1988.

**Model:** We model the log of weekly wages as a function of education, experience, and demographic characteristics.

## 2 Block 1: Setup and Data Exploration

### 2.1 Exercise 0: Setup

Load the packages you will need for today's analysis. The four key packages are **quantreg** (for quantile regression), **AER** (for the CPS1988 dataset), **tidyverse** (for data wrangling and plotting), and **broom** (for tidying model output).

```
# Load packages
```

```
# Load the CPS1988 dataset
```

Now inspect the data. How many rows and columns does it have? What are the variable names and their types?

```
# Inspect the structure of CPS1988
```

```
# Print a summary of all variables
```

#### Questions:

1. How many observations are in the dataset?
  2. Which variables are numeric and which are categorical?
  3. Are there any missing values? (*Hint*: use `is.na()`)
- 

### 2.2 Exercise 1: Explore the Outcome Variable

The outcome we want to model is **weekly wages**. Before fitting any model, explore its distribution.

```
# Plot a histogram of wage on its original scale
```

```
# Now plot a histogram of log(wage)
```

#### Questions:

1. How would you describe the shape of the wage distribution?
2. What happens to the shape when you apply the log transformation?
3. Why is it standard practice to model log wages rather than wages directly?

Now explore the relationship between wages and education, coloured by ethnicity.

```
# Scatterplot of log(wage) vs education, coloured by ethnicity
```

```
# Hint: use geom_point() with colour = ethnicity, then add geom_smooth()
```

#### Questions:

4. Do workers with more education tend to earn more? Does the relationship look linear?
  5. Does the wage-education relationship appear to differ between ethnic groups?
-

## 3 Block 2: OLS Baseline

### 3.1 Exercise 2: Fit an OLS Model

Fit an ordinary least squares regression of log wages on education, experience (including a squared term), ethnicity, and whether the worker lives in a metropolitan area. This gives us our baseline to compare against.

```
# Define the model formula
formula_wages <- log(wage) ~ education + experience + I(experience^2) +
  ethnicity + smsa

# Fit the OLS model

# Print a summary
```

#### Questions:

1. What is the estimated return to an additional year of education? Interpret this coefficient in plain language.
2. How much less do African-American workers earn compared with otherwise similar Caucasian workers? Express this as a percentage.
3. Is the coefficient on  $I(\text{experience}^2)$  positive or negative? What does this imply about the shape of the experience–wage relationship?

Now plot the OLS residuals against the fitted values.

```
# Residuals vs fitted values plot
```

#### Questions:

4. Do the residuals appear to have constant variance (homoscedasticity), or does the spread change with the fitted values?
5. What does this suggest about whether a single mean regression line is adequate?

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## 4 Block 3: Median Regression

### 4.1 Exercise 3: Fit Median Regression ( $\tau = 0.5$ )

Now fit quantile regression at the **median** ( $\tau = 0.5$ ) using `rq()`. Use the same formula as the OLS model.

```
# Fit median regression

# Print a summary with bootstrap standard errors
```

#### Questions:

1. What is the estimated return to education from median regression? Is it higher or lower than the OLS estimate?
2. What does it mean to estimate the *conditional median* rather than the *conditional mean*?
3. Would you expect OLS and median regression to agree more closely on a symmetric distribution or a skewed one? Why?

Now compare the education coefficient from OLS and median regression side by side.

```
# Extract and compare the education coefficients from both models
# Hint: use coef() on each model
```

---

## 5 Block 4: Multiple Quantiles

### 5.1 Exercise 4: Fit Models Across Quantiles

Fit quantile regression at five quantile levels:  $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$ .

```
# Define the quantile levels
taus <- c(0.10, 0.25, 0.50, 0.75, 0.90)

# Fit rq() with tau = taus

# Extract the coefficient matrix
```

Now tabulate the education coefficient across quantile levels.

```
# Extract the education coefficient at each tau
# Hint: use map_dfr() over taus, fitting rq() and pulling the education row
```

#### Questions:

1. Does the return to education increase, decrease, or remain constant as  $\tau$  increases from 0.10 to 0.90?
2. What does this imply about who benefits most from additional years of schooling?
3. The OLS estimate falls somewhere in this range. Does it overestimate or underestimate the return for low earners? For high earners?

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### 5.2 Exercise 5: The Ethnicity Gap Across the Distribution

Now focus on the **ethnicity coefficient** across quantile levels. Extract and display it the same way you did for education.

```
# Extract the ethnicity (African-American) coefficient at each tau
```

#### Questions:

1. Is the wage gap larger or smaller at lower quantiles compared with higher quantiles?
2. What does this imply about *where* in the distribution the wage penalty falls most heavily?
3. The OLS estimate gives one number for the wage gap. Does it capture what is happening at the bottom of the distribution? At the top?
4. What policy implication might follow from this finding?

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## 6 Block 5: Visualisation

### 6.1 Exercise 6: Quantile Lines on a Scatterplot

Plot `log(wage)` against `education` and overlay both the OLS regression line and quantile regression lines at  $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$ . Colour-code the quantile lines.

*Hint:* The quantile lines are straight lines, use `geom_abline()` with the intercept and slope from each fitted model. The OLS line can be added with `geom_smooth(method = "lm")`.

```
# Extract intercept and slope for each tau
# Hint: map_dfr() over taus, rq() on education only (for a clean visualisation)
```

```
# Build the scatterplot with overlaid lines
```

### Questions:

1. What does it mean that the quantile lines are not parallel?
  2. Which quantile line is steepest? Which is shallowest?
  3. How does the fan of lines change as education increases?
- 

## 6.2 Exercise 7: Coefficient Plot Across Quantiles

The **coefficient plot** is the key diagnostic in quantile regression analysis. Plot the education slope coefficient as a function of  $\tau$ , with a 95% bootstrap confidence band and a horizontal reference line for the OLS estimate.

```
# Compute bootstrap confidence intervals at each tau
# Hint: summary(rq_fit, se = "boot", R = 1000)$coefficients

# Build the coefficient plot
```

### Questions:

1. Is the line flat or does it trend upward/downward?
  2. Does the OLS estimate (dashed line) fall within the confidence band at all quantile levels? What does it mean when it does not?
  3. If you could only report one number for the return to education, which would you choose, the OLS estimate, the median regression estimate, or something else? Justify your answer.
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## 7 Block 6: Wrap-Up

*These questions do not require any code. Write your answers directly in the cells below.*

1. In two or three sentences, summarise the main finding about the return to education across the wage distribution.

*Your answer:*

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2. In two or three sentences, summarise the main finding about the ethnicity wage gap.

*Your answer:*

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3. Thinking about your own research: is there a variable in your field where you would expect the effect to differ across the distribution of the outcome? What would the coefficient plot look like if your hypothesis were correct?

*Your answer:*

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4. What is the most important thing you would check before reporting quantile regression results from your own data?

*Your answer:*

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*End of exercises. Well done!*